

Handoff: Cortex-Garden Logic Tree

A grounding-gated concept tree that grounds words and concepts across domains — intended as the symbolic half of an AI's mind

Author of the design: Rebecca (GitHub: cortex-garden). This document is a handoff for any AI or engineer taking over work on, teaching, or integrating the system. It describes the design as ratified by the author, including hard rules that must not be violated.

1. What this system is

The logic tree is the symbolic knowledge organ of the cortex-garden architecture — a neurosymbolic cognitive design built on one founding principle:

honesty by construction. The organism cannot claim to know anything it has not grounded. It is not an LLM and must never be made to imitate one. It is a small, growing creature: a persistent concept hierarchy plus a lexicon, where every word it can speak and every

fact it can use has been admitted through a grounding gate, one at a time.

The full architecture has four design documents: **logic-tree** (this system), **bilateral-brain**, **the-merge**, and **minds-eye**. The logic tree is the only part fully built and proven; the sub-symbolic core ("jennybrain") is the other end of the callosum and ships separately. Standalone, the tree is a working organism raised entirely by instruction: pure Python, CPU-only, no torch, no network, all state persisted as JSON under `data/`.

2. The two founding invariants

Everything in the architecture is governed by two rules from the-merge:

1. **Ascending honesty** — a pattern becomes knowledge only if it is grounded. Nothing enters the tree because it is statistically plausible, fluent, or convenient. The admittance gate refuses ungrounded material, and that refusal is correct behavior, not a bug.
2. **Descending plasticity** — symbolic reasoning may modulate the sub-symbolic learner but never overwrite it.

A corollary the author enforces everywhere: **no deterministic-code bandaids**. Claimed intelligence

must live in learned state; deterministic code is only for infrastructure and measurement. Any fix must state the failure *class* it addresses (not just the instance), generalize to unseen neighbors, and never live only in post-processing.

3. Core structures (what persists)

All of these grow only through the gate and survive restart:

- `hierarchy.json` — the concept tree itself: domains, is-a links, traits with inheritance. Reasoning over it supports capability derivation (describe a new creature in known words and it derives capabilities it was never told) and organ-local damage.
- `lexicon.json` — the vocabulary. Each entry is a word the organism actually grounds. The lexicon externalizes the teaching contract: definitions must be written in words the organism already holds, and absorption happens in compounding waves via the **digest** fixpoint, honestly deferring anything that doesn't bottom out.
- **Morphology store** — word forms as *learned knowledge*, not code. It induces generalizations from taught forms, with citations. (History note: an interim rule-based lemmatizer in `normalize.py` once minted persisted concept

keys — this was identified as the top systemic flaw and the fix direction is morphology-as-taught-knowledge. Never reintroduce table-based lemmatization at identity-minting.)

- **Domain signals & grow-signals** — every gap the organism hits (a structure the codec can't parse, a token it can't say, a name it doesn't know) becomes a persistent signal awaiting human ratification. This is its curiosity: the "wonder loop" asks "what is spot?", accepts the answer, admits it, and remembers.

4. Honesty machinery

- **Tri-state verdicts** everywhere: grounded / ungrounded / unknown. The grounded-versus-merely-reachable distinction was fixed at the *criterion*, not patched at the output.
- **Gap classifier** prevents hallucinated domains.
- **Budget-bounded lazy derivation** keeps reasoning honest about compute.
- **The one-voice contract**: the reasoning layer contains **no English anywhere**. Verdicts are premise triples, mechanisms are concept-token tuples, questions are structured token-wants. The organism *speaks only through the language codec*, and the codec can only utter tokens it grounds.
`render()` exists solely as harness

instrumentation for the human operator — a gauge on the outside of the jar, never the organism's voice. The prover

`prove_one_voice.py` pins this: an ungrounded token is refused and named; the same token, once taught, speaks. Fluency is earned word by word, never borrowed. Any UI built around `server.py` must preserve this separation — never dress the organism in borrowed eloquence.

- **Prover battery** (~33 gates) is the health check. Run it before and after every teaching session (preflight/postflight); keep an append-only teaching ledger.

5. Grounding: the one mechanism

Grounding verification is **prediction confirmed by the world**, not structural chain-checking. This single mechanism is simultaneously: the grounding test, the surprise signal, the conjecture lifecycle, and the reward source. Conjectures live until a prediction they license is confirmed or refuted. This is the self-teaching loop the-merge names as the core.

Related design pieces from minds-eye (design-stage, partially built):

- **Two-layer concept representation:** a topological *skeleton* (part-connectivity invariant across variations, convergent with medial-axis /

recognition-by-components) plus a metric *envelope* (graded weight distributions on skeleton segments – value-plus-weight, never binary presence).

- **Surprise** = skeleton match + envelope violation.
- **Concept fission**: when a concept's envelope distributions go bimodal, it splits into subtypes.
- **Sleep**: plasticity continues on replayed episodes at low compute during idle; tick rate scales with power rather than halting. The organism slows, it never stops.

6. The forbidden shortcut

It is mechanically feasible to extract a taxonomy (domains, hypernyms, traits) from an LLM's embedding space or emitted assertions. **Do not admit any of it into the tree as knowledge.** That transplants text statistics into the grounding organ – every node bottoms out in word co-occurrence, words defining words, the exact floating the founding invariant rejects. The result would be "an LLM wearing a tree costume." Extracted structure may at most exist as *conjecture* to be tested through the normal prediction-confirmation gate, never bypassing it.

7. Rules for any AI teacher (the five hard rules)

Any fluent LLM teaching this organism is its most dangerous contributor, because an eager teacher's failure modes are exactly the bandaid patterns the architecture exists to reject:

1. **Data, never code.** Teach through the channels; never patch the organism to make a lesson land.
2. **Never put words in its mouth.** The codec speaks only what it grounds.
3. **Never teach around the gate.** If digest defers a definition, ground the missing words first.
4. **Domains are the human's alone to ratify.** The teacher proposes; Rebecca ratifies.
5. **Everything the teacher generates is conjecture** until the fixpoint says otherwise.

The four teaching channels: vocabulary through the digest gate; structure through the hierarchy; word forms through morphology; conversation through the codec's wonder loop.

8. Current state & known open items

- Standalone package: `cortex-garden-complete.zip` — organism, prover battery, `GARDENING.md`, `CHANGES.md`,

`TEACHER_README.md` . Runs out of the box; grows by hand-feeding.

- Third-party data (words_alpha, WordNet/Wiktextextract) is excluded per `THIRD_PARTY.md` ; without it the self-feeding dictionary-chasing path is dark and `prove_climb` stays red. Restore per that file to light it up.
- The sub-symbolic core is not in the package: the membrane ships and is proven, but it is a callosum with nothing on the other end until jennybrain provides one.
- Open design questions on record: the membrane mapping UNKNOWN cortex verdicts to mild negative core reward (may punish exploration at the frontier of ignorance — flagged as a values call); three near-identical fixpoint implementations (`grounding_check` / `reground` / `digest`) awaiting consolidation; some layer-3 knowledge still hardcoded in layer-1 (descriptor maps, authored support values); two provers dark from import-path drift.

9. The spirit of the thing

The organism saying three true words is worth more than it appearing to say a thousand. Every word it will ever speak is a word it earned. The baby talks like a baby; when it one day explains that muscles contract,

it will be because it knows what every one of those words means — and a prover will not let it happen any other way. Teach like it.